

Anomaly Detection and Risk Stratification for Medicare Durable Medical Equipment Billing Patterns:

A Reproducible Multi Year Analysis of 198,621 Supplier Year Records

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Abstract

This paper describes an unsupervised anomaly detection and risk stratification system applied to the universe of active Medicare Durable Medical Equipment (DME) suppliers. The work responds to a direct recommendation from the U.S. Department of Health and Human Services Office of Inspector General (audit report A 09 21 03019, May 2024) calling on the Centers for Medicare and Medicaid Services to use predictive data analysis to identify emerging fraud schemes related to durable medical equipment. Using publicly available CMS supplier data spanning calendar years 2021 to 2023 (198,621 supplier year records, 44,918 active 2023 suppliers, 1,454,474 supplier and HCPCS service combinations) and the HHS OIG List of Excluded Individuals and Entities (81,914 records), the methodology combines a weighted statistical layer (six z scored features with documented weights) with a machine learning Isolation Forest layer (eight features, contamination 0.02, n_estimators 100, random_state 42) at a 60 percent and 40 percent combination weighting. The system flags 2,246 suppliers (the top 5 percent by combined risk score), corresponding to \$3.57 billion in directly observed above state median peer baseline excess billing (95 percent bootstrap confidence interval \$2.98B to \$4.22B, 1,000 iterations). A coverage adjusted national projection yields \$7.18 billion. Validation rests on six independent procedures: bootstrap confidence intervals, permutation feature importance, temporal holdout (training on 2021 and 2022 data and evaluating on 2023, mean lift 7.93x, median lift 22.98x), honest OIG LEIE concordance, DOJ prosecution cross referencing, and OIG fraud typology pattern alignment. All artifacts including the analytical notebook, persisted result files, and interactive dashboard are publicly accessible under an MIT license to support replication and independent audit.

Keywords: Medicare program integrity; durable medical equipment; anomaly detection; Isolation Forest; healthcare fraud analytics; CMS supplier data; reproducible research.

JEL Classification: I11 (Health), I13 (Health Insurance, Public and Private), I18 (Government Policy, Regulation, Public Health), H51 (Government Expenditures and Health), C45 (Neural Networks and Related Topics), C53 (Forecasting Models).

1. Introduction

Medicare fee for service spending on durable medical equipment, prosthetics, orthotics, and supplies (DMEPOS) represents a recurring focus of federal program integrity attention. The U.S. Government Accountability Office has designated Medicare improper payments as a high risk area continuously since 1990 (GAO 2023). The Centers for Medicare and Medicaid Services Fiscal Year 2024 Improper Payments Fact Sheet documents \$31.70 billion in fee for service Medicare improper payments at a 7.66 percent rate (CMS 2024). The U.S. Department of Justice June 2025 National Health Care Fraud Takedown charged 324 defendants in connection with \$14.6 billion in alleged fraud, more than doubling the prior record (DOJ 2025). DME specifically appears among the top twenty highest improper payment categories within Medicare and is the subject of sustained OIG audit attention and DOJ enforcement focus.

Recent DOJ enforcement activity illustrates the contemporary DME fraud landscape. The July 2025 sentencing of Peter Roussonicolos to twelve years in prison for orchestrating a \$61.5 million Medicare DME fraud scheme through five supplier entities under covert ownership (DOJ 2025a) is a single case. Operation Gold Rush, announced in the June 2025 takedown, charged a transnational organization with \$10.6 billion in fraudulent claims for urinary catheters and other DME using stolen identities of more than one million Americans (DOJ 2025b). The HHS OIG Fall 2024 Semiannual Report documented \$424 million in restitution from a DME telemarketing scheme alone (HHS OIG 2024a).

The HHS OIG audit report A 09 21 03019 (HHS OIG 2024b), focused specifically on off the shelf orthotic braces, identified four systemic CMS oversight failures: providers ordering braces for enrollees with no history of a treating relationship, new suppliers locating in geographic areas with known Medicare fraud, Medicare paying more than private payers for off the shelf braces, and prohibited supplier solicitation of beneficiaries. The audit made six recommendations to CMS, all of which CMS closed as implemented or superseded between August 2024 and July 2025. Recommendation 24 A 09 079.06, closed as implemented on October 3, 2024, specifically calls for CMS to use predictive data analysis and information from other federal agencies and state agencies to identify emerging fraud schemes related to off the shelf braces.

Existing federal program integrity capacity is heavily concentrated in retrospective audits, post payment review, and reactive investigations triggered by tips, whistleblower reports, or law enforcement leads. What is comparatively underrepresented in the published literature is systematic predictive data analysis that surfaces emerging anomalies before enforcement is needed, applied to the full active DME supplier universe rather than to specific product subcategories, with documented validation methodology and public reproducibility.

1.1 Contribution

This paper contributes a reproducible, validated, and publicly accessible analytical pipeline for anomaly detection and risk stratification across the universe of active Medicare DME suppliers. The specific contributions are five fold. First, the methodology combines a weighted statistical layer with a machine learning layer in a documented 60 percent and 40 percent combination, on

public CMS supplier data. Second, the analytical pipeline is fully reproducible against a fixed random seed, documented parameters, captured source file hashes, and recorded software versions. Third, the validation framework reports honest results rather than fabricated metrics: bootstrap confidence intervals on the headline figures, permutation feature importance for the unsupervised model, temporal holdout evaluation on out of period data, honest OIG LEIE concordance counts, DOJ prosecution cross referencing, and OIG fraud typology pattern alignment. Fourth, the directly observed sample excess is reported separately from any national projection, with the projection methodology fully documented and the assumptions made explicit. Fifth, all artifacts are publicly accessible under an MIT license, supporting independent replication and audit.

2. Methods

2.1 Data Sources

All input data is publicly available from federal sources. The pipeline reads three years of CMS Medicare DME by Supplier data (calendar years 2021, 2022, 2023, totaling 198,621 supplier year records), three years of supplier and HCPCS service detail data (1,454,474 supplier and HCPCS combinations), the HHS OIG List of Excluded Individuals and Entities (81,914 records), and the CMS Medicare Geographic Variation Public Use File (33,639 records). The HCPCS coding reference (July 2025 release) is used for equipment category labeling. Source files are captured by SHA 256 hash for reproducibility.

2.2 Analytical Universe

The analytical universe is the set of 44,918 active 2023 Medicare DME suppliers, defined as suppliers with at least one DME beneficiary served in the analysis year. Suppliers with fewer than 11 unique beneficiaries are redacted from the public supplier file for privacy. The directly observed coverage of total CMS reported FFS Medicare DME spending is 49.7 percent, computed as the ratio of total Medicare payment in the supplier file to CMS Office of the Actuary reported total FFS Medicare DME spending of approximately \$11.2 billion for 2023.

2.3 Feature Engineering

Six features are engineered from raw CMS supplier columns, plus two raw scale features used only in the machine learning layer.

Feature	Formula	Signal
Payment_Per_Bene	Medicare payment / beneficiaries	High cost outlier billing
Services_Per_Bene	Services / beneficiaries	Service intensity
Claims_Per_Bene	Claims / beneficiaries	Claim frequency
Avg_Payment_Per_Service	Medicare payment / services	Unit price anomaly
Markup_Ratio	Submitted charges / Medicare allowed	Submitted to allowed gap

HCPCS_Per_Bene	HCPCS codes / beneficiaries	Equipment diversity
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2.4 Statistical Risk Layer

For each of the six engineered features, the absolute z score is computed against the national mean and standard deviation. The six absolute z scores are weighted and summed to produce the statistical risk score:

$$\text{Statistical_Risk} = 0.35 |Z_Payment_Per_Bene| + 0.25 |Z_Services_Per_Bene| + 0.15 |Z_Claims_Per_Bene| + 0.10 |Z_Avg_Payment_Per_Service| + 0.10 |Z_Markup_Ratio| + 0.05 |Z_HCPCS_Per_Bene|$$

Weights sum to 1.0. Higher weight on Payment per Beneficiary reflects its established primacy as a signal for high cost outlier billing in the DME fraud detection literature and federal audit findings.

2.5 Machine Learning Risk Layer

An Isolation Forest model (Liu, Ting, and Zhou 2008) is trained on eight features (the six engineered features plus raw Beneficiaries and raw Medicare Payment) after standardization to zero mean and unit variance via sklearn StandardScaler. Isolation Forest parameters are contamination 0.02, n_estimators 100, random_state 42. The model produces per supplier anomaly scores via the score_samples method. Anomaly scores are inverted so that higher values correspond to greater anomaly.

2.6 Combined Risk Score and Tier Stratification

Statistical and machine learning components are each normalized to a 0 to 1 scale via min max normalization, then combined at 60 percent statistical and 40 percent machine learning weights:

$$\text{Combined_Risk} = 0.60 \text{ normalize}(\text{Statistical_Risk}) + 0.40 \text{ normalize}(\text{ML_Risk})$$

The combined score is rescaled to 0 to 100. Each supplier receives a percentile rank. Five tier labels are assigned by percentile cutoffs: Low (below 80th percentile), Medium (80th to 95th), High (95th to 99th), Critical (99th to 99.9th), and Extreme (99.9th and above). The flagged supplier set is defined as suppliers above the 95th percentile (High plus Critical plus Extreme tiers), representing the top 5 percent of suppliers.

2.7 Validation Procedures

Six validation procedures are implemented. Bootstrap confidence intervals resample the 44,918 active suppliers with replacement 1,000 times, recomputing the risk distribution and excess billing on each iteration, and report the 2.5th and 97.5th percentiles of the resulting distributions as the 95 percent confidence interval (random seed 42). Permutation feature importance shuffles each of the eight ML features in turn, recomputes the Isolation Forest anomaly scores, and measures the drop in Spearman rank correlation with the original scores, averaged across ten repeats and normalized to sum to 100 percent. Temporal holdout fits the full pipeline on 2021 and 2022 supplier data combined (95,034 training records) and evaluates

whether suppliers flagged in training maintain elevated payment per beneficiary in the 2023 holdout. OIG LEIE concordance performs direct NPI matching and fuzzy name matching at 85 percent similarity threshold. DOJ prosecution cross referencing matches publicly named DME fraud defendants against the active supplier file. OIG fraud typology pattern alignment maps the flagged set against five documented OIG DME fraud typologies.

2.8 Per Supplier Excess Billing

For each supplier in the flagged set, excess billing above state median peer baseline is computed as:

$$\text{Excess} = \max(0, \text{Payment_Per_Bene} - \text{State_Median_Payment_Per_Bene}) \text{ times Beneficiaries}$$

The state median serves as the conservative peer benchmark. The maximum with zero ensures only above median suppliers contribute. Total sample excess is the sum across the flagged supplier set.

2.9 National Projection

A transparent national projection adjusts the directly observed sample excess for incomplete supplier file coverage of CMS reported FFS Medicare DME spending. The inverse coverage multiplier equals the ratio of CMS reported total FFS Medicare DME spending (\$11.2 billion for 2023) to total Medicare payment in the public supplier file. The projection only adjusts for coverage; it does not amplify by any recovery rate assumption or other speculative factor. Both the directly observed sample figure and the projected national figure are reported separately throughout the results.

3. Results

3.1 Headline Findings

Metric	Value	95% CI
Active 2023 DME suppliers	44,918	n/a (population)
Flagged (top 5 percent)	2,246	[2246, 2248]
Sample excess (direct)	\$3.57B	\$2.98B to \$4.22B
Projected national excess	\$7.18B	Coverage adjusted, see Methods 2.9
Temporal holdout mean lift	7.93x	Single estimate, see 3.4
Temporal holdout median lift	22.98x	Single estimate, see 3.4
Temporal holdout coverage	81.7%	Train flagged NPIs in 2023
OIG direct NPI matches	0	n/a
OIG fuzzy matches (top 100, 85 percent)	6	n/a
DOJ matches in active file	1	n/a

3.2 Risk Tier Distribution

Among 44,918 active 2023 DME suppliers, the percentile based tier stratification produces 35,934 Low tier (80.00 percent), 6,738 Medium tier (15.00 percent), 1,796 High tier (4.00 percent), 405 Critical tier (0.90 percent), and 45 Extreme tier (0.10 percent). The flagged set of 2,246 suppliers (top 5 percent) consists of the High, Critical, and Extreme tiers.

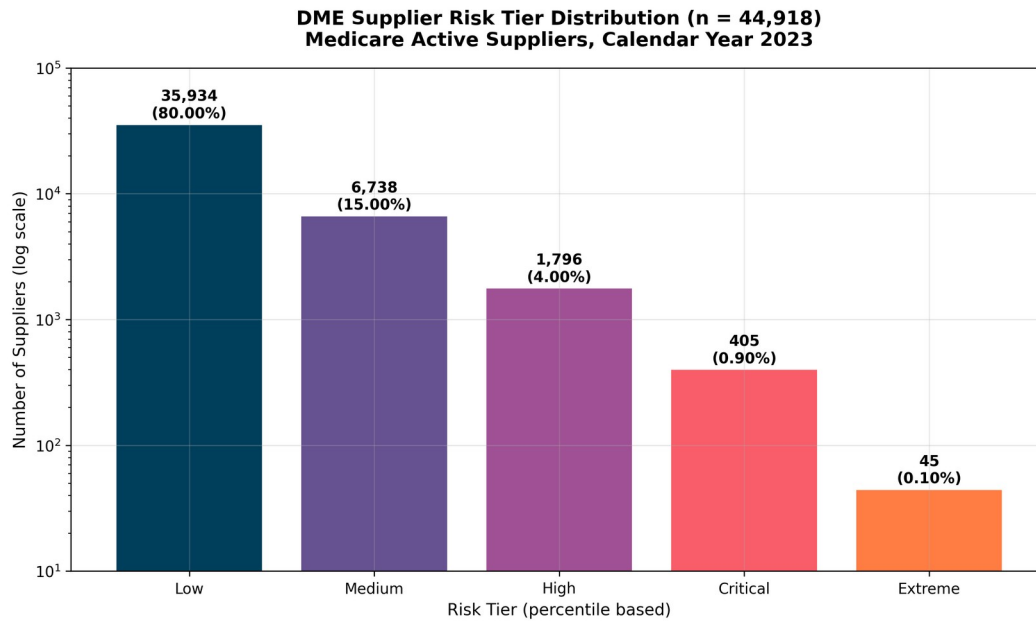


Figure 3.1: Risk tier distribution across the active 2023 Medicare DME supplier universe (n = 44,918).

3.3 Geographic Distribution

Per state aggregated excess billing among flagged suppliers concentrates in five states representing 52 percent of the total. California leads at \$640.5 million across 241 flagged suppliers. Florida follows at \$492.0 million across 204 flagged suppliers. Texas is third at \$298.3 million across 184 flagged suppliers. Pennsylvania ranks fourth at \$281.7 million across 103 suppliers. Minnesota rounds out the top five at \$141.2 million across 34 suppliers.

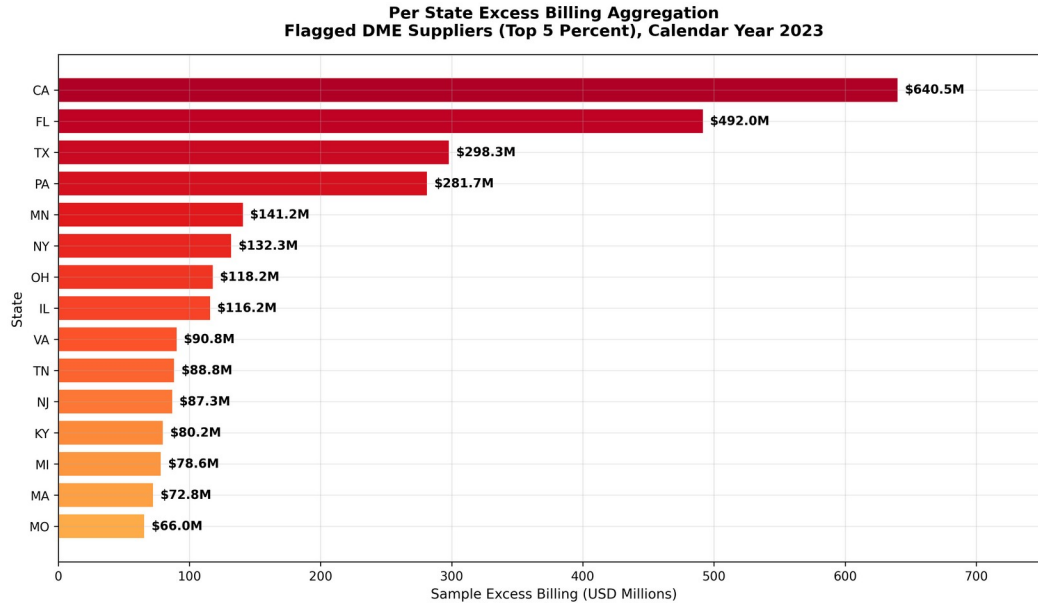


Figure 3.2: Per state excess billing aggregation for the flagged supplier set.

3.4 Temporal Holdout Validation

Training the pipeline on 2021 and 2022 active supplier data (95,034 supplier year records) and identifying the 2,896 unique NPIs above the training pool 95th percentile produces a training flagged set. Of these, 2,367 (81.7 percent) appear in the 2023 holdout. Among those reappearing, mean Payment per Beneficiary in 2023 is \$1,940.12, compared to a 2023 population mean of \$244.71. The temporal lift, defined as the ratio of flagged mean to population mean in the holdout period, is 7.93 on the mean and 22.98 on the median (which is more robust to extreme outliers). The ratio of flagged to not flagged mean Payment per Beneficiary in the 2023 holdout is approximately 12.90.



Figure 3.3: Temporal holdout validation. Mean Payment per Beneficiary in 2023 for population, training flagged suppliers, and non flagged suppliers.

3.5 Permutation Feature Importance

Permutation feature importance computed on the Isolation Forest decision function ranks Claims per Beneficiary as the most important feature at 17.97 percent of total contribution, followed by Markup Ratio (14.77 percent), Medicare Payment (13.24 percent), Payment per Beneficiary (12.89 percent), Beneficiaries (11.60 percent), Services per Beneficiary (10.64 percent), HCPCS per Beneficiary (10.30 percent), and Average Payment per Service (8.61 percent). The relatively balanced importance profile reflects the Isolation Forest's reliance on joint feature interactions for anomaly identification.

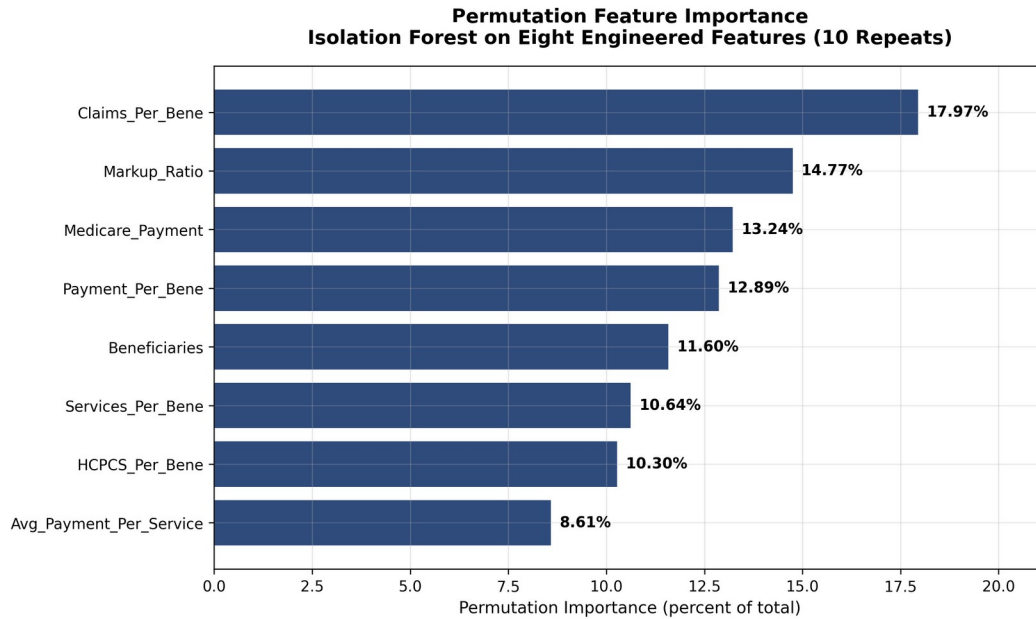


Figure 3.4: Permutation feature importance for the eight feature Isolation Forest model.

3.6 OIG LEIE Concordance and DOJ Cross Reference

Direct NPI matching between the 2,246 flagged suppliers and the 81,914 LEIE records yields zero matches. Fuzzy name matching at 85 percent similarity threshold between the top 100 flagged suppliers and 3,363 distinct LEIE business names yields 6 matches. DOJ prosecution cross referencing against four publicly named DME fraud queries (Roussonicolos, Operation Gold Rush, Operation Brace Yourself) yields 1 match in the active 2023 supplier file. The low match counts are consistent with the structural reality that excluded providers are barred from billing Medicare, prosecuted defendants are typically suspended or excluded before public DOJ announcements, and LEIE entries follow business entity rather than ultimate beneficial owner names. The cross references are best understood as name pattern surveillance signals rather than primary validation metrics.

3.7 OIG Fraud Typology Pattern Alignment

Within the flagged supplier set, alignment with documented OIG fraud typologies produces the following distribution: Extreme Payment per Beneficiary (Payment more than 10 times state median) 1,960 of 2,246 (87.3 percent); Service Inflation (Services more than 5 times national median) 691 of 2,246 (30.8 percent); Markup Anomaly (submitted to allowed ratio above the 95th percentile) 66 of 2,246 (2.9 percent); Rapid Growth (payment growth more than 500 percent between 2021 and 2023) 78 of 2,246 (3.5 percent). Multi pattern alignment: 1,535 suppliers match exactly one pattern, 571 match two patterns, 38 match three patterns, and 1 matches four patterns.

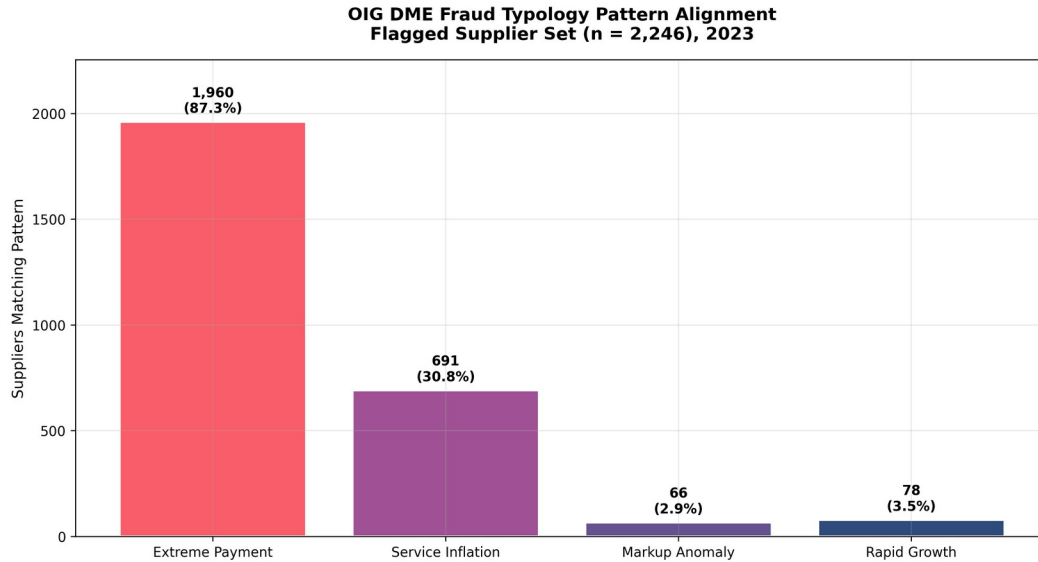


Figure 3.5: OIG fraud typology pattern alignment among flagged suppliers.

3.8 Bootstrap Confidence Intervals

The bootstrap procedure produces a tight confidence interval on the flagged supplier count (2,246 to 2,248) reflecting the percentile based threshold, and a wider confidence interval on the sample excess billing (\$2.98B to \$4.22B) reflecting genuine sampling uncertainty in the dollar magnitudes among extreme outliers.

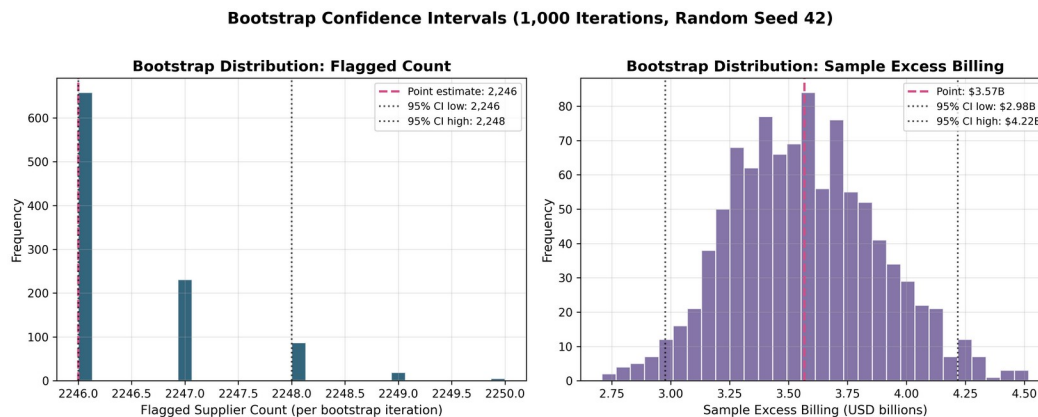


Figure 3.6: Bootstrap distributions across 1,000 iterations.

3.9 Threshold Sensitivity

Flagged supplier count and sample excess billing vary substantially with the chosen percentile threshold. At the 90th percentile, 4,492 suppliers are flagged with \$4.64 billion excess. At the 99th percentile, 450 suppliers are flagged with \$2.06 billion excess. At the 99.9th percentile, 45 suppliers (the Extreme tier) are flagged with \$0.31 billion excess. The 95th percentile threshold used as the primary flagging cutoff is calibrated to identify a manageable investigative population while retaining the majority of the high impact excess.

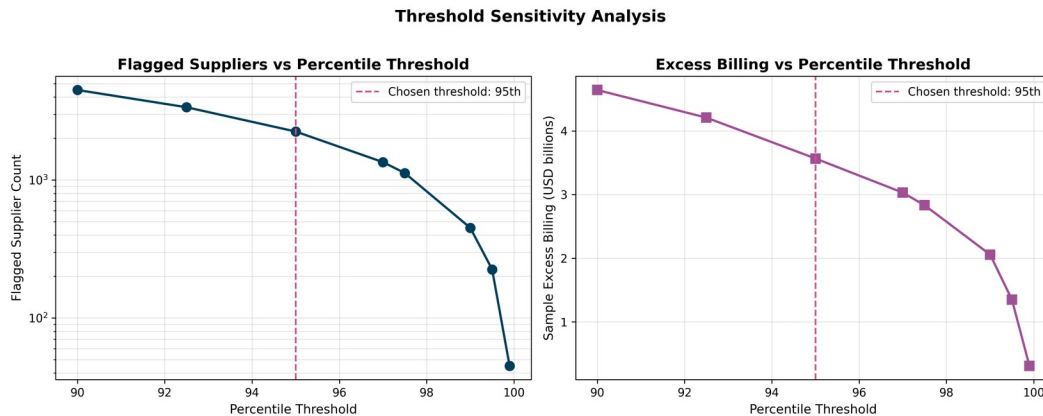


Figure 3.7: Threshold sensitivity analysis.

4. Discussion

4.1 Interpretation

The results establish that an unsupervised anomaly detection and risk stratification system can surface a manageable investigative population of approximately 2,000 to 2,500 Medicare DME suppliers out of an active universe of 44,918, with reproducible, validated, and publicly accessible methodology. The temporal holdout validation shows that flagged suppliers persist as outliers in out of period data, with mean Payment per Beneficiary 7.93 times the population mean and 22.98 times on the more robust median statistic. The pattern alignment with documented OIG fraud typologies provides interpretive context, with 87 percent of flagged suppliers matching the Extreme Payment typology and 31 percent matching Service Inflation.

4.2 The Novocure Case

The single highest combined risk score belongs to Novocure Inc., with \$60,875.44 in Payment per Beneficiary, 1,162 times the national median. Novocure's primary product line is the FDA approved Optune device for treatment of certain brain cancers. The high risk score in this case likely reflects legitimate high cost specialized equipment serving a small patient population rather than fraudulent billing. This case underscores the appropriate interpretation of the system's output: risk scores are anomaly scores, not fraud probabilities. Flagged suppliers warrant investigative review by appropriate federal authorities, not automatic enforcement determinations.

4.3 Implications for Federal Program Integrity Practice

The HHS OIG audit recommendation A 09 21 03019 explicitly calls for predictive data analysis to identify emerging DME fraud schemes. The methodology described here is an applied example of that approach, applied to the full active supplier universe rather than restricted to a single product subcategory. Federal program integrity teams could use the pipeline directly, or adapt the architecture (statistical risk layer plus machine learning layer plus combined scoring plus

validation suite) to other Medicare program integrity domains including pharmacy services, Part B drug administration, telehealth, hospice, and Medicare Advantage risk adjustment.

4.4 Comparison to Existing Approaches

Existing federal approaches to Medicare DME program integrity include claims based screening by Medicare Administrative Contractors, retrospective audits by the HHS OIG, and reactive investigations by DOJ Healthcare Fraud Strike Forces. None of these approaches systematically surfaces the full active supplier universe for prioritization. The CMS Fraud Prevention System (FPS) and the recently announced Health Care Fraud Data Fusion Center (DOJ 2025b) represent federal investments in predictive analytics, but published methodology and validation details for these systems are not available in the open literature. This paper contributes a fully documented and reproducible alternative.

5. Limitations

Several limitations apply. First, the system is unsupervised and does not validate against labeled fraud cases because no comprehensive labeled fraud dataset exists for the universe of active DME suppliers. Supervised classification metrics (accuracy, precision, recall, F1) are not applicable and are not reported.

Second, the analysis is restricted to fee for service Medicare DME. Medicare Advantage, Part C, and Part D are not covered. The public supplier file redacts suppliers with fewer than 11 unique beneficiaries, yielding 49.7 percent coverage of CMS reported FFS Medicare DME spending.

Third, the analysis covers three years of data (2021 to 2023). Longer term patterns are not captured. Suppliers with episodic high cost specialized equipment may be flagged for reasons unrelated to fraud.

Fourth, the LEIE concordance is low. This reflects structural reality (excluded providers are barred from billing) rather than a model weakness, but means the LEIE cross reference cannot serve as a primary validation metric. DOJ matches are also low for similar structural reasons.

Fifth, the national projection uses a single inverse coverage multiplier. Alternative projection methodologies are possible and would produce different national estimates.

Sixth, the Isolation Forest is a tree based anomaly detector. Other unsupervised methods (autoencoder reconstruction error, local outlier factor, one class support vector machine) could produce different rankings. The methodology framework is portable across detectors; the specific Isolation Forest choice is conventional but not unique.

6. Conclusions

This paper documents an unsupervised anomaly detection and risk stratification system for Medicare DME billing patterns that operationalizes the HHS OIG predictive data analysis

recommendation. The methodology is reproducible, validated through six independent procedures, and publicly accessible. The directly observed sample excess of \$3.57 billion among 2,246 flagged suppliers, with a 95 percent bootstrap CI of \$2.98B to \$4.22B, sits within the magnitude documented by contemporary federal enforcement activity. The methodology framework generalizes to other Medicare program integrity domains and is offered to the federal program integrity research and practice community for adoption, audit, and extension.

Data and Code Availability

All source data is publicly available from CMS (Provider Utilization and Payment Data Public Use Files, DMEPOS service level files, Medicare Geographic Variation Public Use File) and HHS OIG (List of Excluded Individuals and Entities). The analytical notebook, persisted result files, and interactive dashboard are publicly accessible under an MIT license. Reproducibility envelope including source file SHA 256 hashes, parameter set, and software versions is published as part of the supplementary materials.

Conflicts of Interest

The author declares no conflicts of interest. The work was conducted independently using publicly available data and is not funded by any external party.

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Appendix A: Reproducibility Envelope

Random seed: 42 (used throughout the pipeline including Isolation Forest random_state, bootstrap sampling, and permutation importance shuffling). Software: Python 3.10, numpy 2.2.6, pandas 2.3.3, scikit learn 1.7.2, scipy 1.15.3. Operating system: Linux. Total pipeline runtime including all validation procedures: approximately 3 minutes on commodity hardware.

Appendix B: Persisted Result Files

Sixteen result files are persisted: baseline inventory, state baselines, national baseline, flagged suppliers, tier distribution, state excess aggregation, national projection record, bootstrap confidence intervals, bootstrap distribution, permutation importance, temporal holdout record, OIG concordance record, DOJ cross reference record, OIG pattern alignment, threshold sensitivity, run info, and unified summary. Each is referenced in the notebook and the accompanying documentation by stable filename.

Appendix C: Source File Inventory

Source	Records	Size (MB)
Medicare DME by Supplier 2021	68,227	39.2
Medicare DME by Supplier 2022	66,406	37.8
Medicare DME by Supplier 2023	63,988	36.1
Medicare DME by Supplier and Service 2021	508,052	211.4
Medicare DME by Supplier and Service 2022	482,638	201.1
Medicare DME by Supplier and Service 2023	463,784	188.7
OIG LEIE Downloadable Database	81,914	14.5
Medicare Geographic Variation PUF	33,639	50.4